OPEN ELECTIVE • SEMESTER 6 • ELECTRONICS / ECE

Introduction to Hybrid and Electric Vehicles

A full-screen handbook for course code **6042C**. It converts the official syllabus into a usable study guide with concepts, diagrams, equations, solved examples, quick revision, Malayalam support notes and exam practice. This is structured as a handbook for understanding EV systems from vehicle motion to charging station level.

COURSE
6042C

CREDITS
4

PERIODS
60

CATEGORY
Open Elective

Syllabus Structure Converted to Study Plan

The subject has four outcomes: EV and HEV technologies, energy efficiency of drive trains, power and energy need of EV/HEV, and energy management strategies. The handbook follows the same order so exam preparation stays aligned.

Course Objectives

- ✓ Design and develop basic schemes of electric and hybrid electric vehicles.
- ✓ Choose suitable drive scheme based on application and resources.
- ✓ Understand EV motors, batteries, charging, smart charging and EV trends.

> EV □□□□□□□ battery □□□□□□□ □□□□. Battery, converter, motor, mechanical load, controller, CAN communication, charging station □□□□□□□ □□□□□□□ complete system □□□.

CO and Hour Distribution

CO	Outcome	Hours	Level
CO1	Extend EV and HEV technologies and challenges	14	Understanding
CO2	Evaluate energy efficiency of vehicle drive trains	20	Understanding
CO3	Evaluate power and energy need of HEV/EV	14	Understanding
CO4	Illustrate energy management strategies	10	Understanding

Module Roadmap

Module	Core content	What to master
M1	History, EV types, vehicle dynamics, HEV introduction and importance.	Classify EV types, derive road-load equations, explain why EVs matter.
M2	Electric drive train topologies, components, DC/IM/PM/SRM drives, HEV topologies.	Draw topologies, explain power flow, compare motor drive choices.
M3	Energy storage, battery types, fuel cell, supercapacitor, flywheel, hybrid storage, drive sizing.	Compare storage systems, calculate ratings, select motor and battery capacity.

M4	Vehicle communication, CAN, energy management strategies, charging station schematic.	Explain CAN role, classify EMS, draw charging station block diagram.

Module 1 — EV and Hybrid Vehicle Foundation

Need, history, principle of operation, EV types, vehicle motion equations, forces, social and environmental importance.

Need for Electric Vehicles

Transportation consumes large fuel energy and creates urban emissions. EVs reduce tail-pipe emissions, allow regenerative braking, use efficient electric motors and can use renewable-grid energy. The real benefit depends on electricity source, battery life, charging infrastructure and recycling.

EV Types

- **BEV:** Battery electric vehicle, no ICE propulsion.
- **HEV:** ICE and electric machine share propulsion and energy recovery.
- **PHEV:** HEV with larger battery and external charging.
- **FCEV:** Fuel cell produces electricity from hydrogen.
- **Solar vehicle:** Solar panels assist charging; practical power is limited by area and sunlight.

Hybrid Vehicle Meaning

A hybrid vehicle uses at least two energy sources or converters. Normal HEV uses chemical fuel through ICE and electrochemical battery through motor. Hybridization allows engine stop-start, regenerative braking, peak torque assist and efficient operating-point control.

Social and Economic Importance

EVs reduce local pollution, reduce oil dependence, create battery/charging industries and support renewable energy integration. Challenges include high initial cost, battery raw materials, range anxiety, recycling and charging infrastructure.

Vehicle Dynamic Equations

The traction motor must overcome road-load forces. In design problems, first calculate force, then power, then motor/battery requirement.

Total tractive force: $F_t = F_{rr} + F_{ad} + F_g + F_a$

Rolling resistance: $F_{rr} = C_{rr}\ m\ g\ \cos\theta$

Aerodynamic drag: $F_{ad} = 0.5\ \rho\ C_d\ A\ v^2$

Gradient force: $F_g = m\ g\ \sin\theta$

Acceleration force: $F_a = m\ a$

Wheel power: $P = F_t \times v$

Common mistake: Students calculate force but forget velocity. Motor power is not force alone; power is force multiplied by speed.

Mini Solved Example

Problem: A 1000 kg EV travels at 20 m/s on level road. Crr = 0.015, ρ = 1.2 kg/m³, Cd = 0.30, frontal area = 2.2 m². Acceleration is zero. Find wheel power.

1. Rolling force = 0.015 × 1000 × 9.81 = 147.15 N.
2. Aero drag = 0.5 × 1.2 × 0.30 × 2.2 × 20² = 158.4 N.
3. Total force = 305.55 N.
4. Power = 305.55 × 20 = 6111 W ≈ 6.1 kW.

Module 2 — Electric Drive Trains and Propulsion Units

Electric traction, drive-train topologies, power flow control, efficiency analysis, electric propulsion components and motor-drive comparison.

Electric Drive Train Blocks

- ✓ Energy source: battery, fuel cell or supercapacitor.
- ✓ Power converter: DC-DC converter, inverter and charger.

- ✓ Traction motor: DC motor, induction motor, PMSM/BLDC or SRM.
- ✓ Transmission: fixed gear, differential or wheel drive.
- ✓ Control system: accelerator input, motor controller, BMS, protection and communication.

Power Flow Modes

Mode	Power path	Purpose
Motoring	Battery → inverter → motor → wheels	Propulsion
Regeneration	Wheels → motor → inverter → battery	Recover braking energy
Charging	Grid/charger → battery	Replenish energy
Hybrid assist	Battery + ICE → wheels	Peak torque and fuel saving

Motor Drive Comparison

Motor	Merits	Limitations	Use in EV
DC motor	Simple speed control, high starting torque.	Brush maintenance, lower reliability.	Older/small EVs and educational models.
Induction motor	Robust, no magnets, good high-speed capability.	Needs inverter and vector control.	Rugged traction drives.
Permanent magnet motor	High efficiency and power density.	Magnet cost and temperature sensitivity.	Common in modern EV traction.
Switched reluctance motor	Simple rotor and high temperature capability.	Torque ripple and acoustic noise.	Emerging EV applications.

Hybrid Drive-Train Topologies

Series Hybrid

ICE drives generator; generator supplies motor/battery; wheels are driven only by motor. Mechanically simple, but multiple energy conversions create loss.

Parallel Hybrid

ICE and motor can both drive wheels. Efficient for highway operation, but mechanical coupling is more complex.

Series-Parallel Hybrid

Combines series and parallel benefits using power-split mechanism. Control is advanced but energy optimization is better.

Plug-in Hybrid

Large battery can be externally charged. It gives electric-only range and ICE backup for long trips.

> Hybrid topology answer □□□□□□□□□□ source → converter → motor → wheel power flow arrow must be clear.

Efficiency Chain

$$\eta_{overall} = \eta_{battery} \times \eta_{converter} \times \eta_{motor} \times \eta_{gear}$$

If battery efficiency is 0.95, inverter efficiency 0.96, motor efficiency 0.92 and gear efficiency 0.97, overall efficiency = 0.95 × 0.96 × 0.92 × 0.97 = 0.814 ≈ 81.4%.

Module 3 — Energy Storage and Drive System Sizing

Battery parameters, chemistry comparison, fuel cells, supercapacitors, flywheel, hybrid storage and rating calculation.

Battery Parameters

- ✓ **Voltage:** decides inverter and motor drive level.
- ✓ **Capacity Ah:** amount of charge available.
- ✓ **Energy Wh:** voltage × Ah.

- ✓ **SOC:** remaining battery percentage.
- ✓ **SOH:** ageing condition.
- ✓ **C-rate:** charge/discharge rate relative to capacity.
- ✓ **Specific energy:** Wh/kg, important for range.
- ✓ **Specific power:** W/kg, important for acceleration.

Storage Comparison

Storage	Strength	Weakness
Lead acid	Low cost, mature.	Heavy, low energy density.
Nickel based	Better energy than lead acid.	Memory effect/cost issues.
Lithium ion	High energy density and efficiency.	Needs BMS and thermal protection.
Supercapacitor	High power and fast charge.	Low energy density.
Fuel cell	Long range and fast refuel potential.	Hydrogen infrastructure and cost.

Drive Sizing Method

1. Fix vehicle mass, speed, acceleration, gradient and range target.
2. Calculate rolling, aerodynamic, gradient and acceleration forces.
3. Calculate wheel power and motor shaft power after drivetrain loss.
4. Select motor peak power and continuous power.
5. Select inverter current, voltage and thermal rating.
6. Calculate battery energy from range and Wh/km consumption.
7. Add reserve for ageing, temperature, depth of discharge and safety margin.

Key Sizing Equations

$$\text{Battery energy (Wh)} = \text{Nominal voltage} \times \text{Ah}$$

Required capacity (Ah) = Required Wh / Pack voltage

Range (km) = Usable Wh / Wh per km

Motor torque $T = P / \omega$

Wheel torque = Tractive force × wheel radius

Mini Solved Example — Battery Pack Capacity

Problem: An EV consumes 120 Wh/km and must travel 80 km. Pack voltage is 300 V. Assume usable depth is 80%. Find required nominal battery capacity.

1. Usable energy required = 120 × 80 = 9600 Wh.
2. Nominal energy = 9600 / 0.8 = 12000 Wh.
3. Capacity Ah = 12000 / 300 = 40 Ah.

Answer: A 300 V, 40 Ah pack is the minimum nominal capacity before adding design margin.

Module 4 — Communication, Energy Management and Charging

Vehicle networks, CAN, energy management strategies, implementation issues and charging station schematic.

Why CAN is Used in EV

EV control units must exchange messages reliably: BMS sends battery voltage/current/SOC, motor controller sends speed/torque/fault, charger sends charging status, and vehicle control unit coordinates torque and safety limits. CAN is robust, priority-based and supports distributed control.

- ✓ Two-wire differential communication.
- ✓ Message-based arbitration.
- ✓ High noise immunity.
- ✓ Fault detection and error handling.

✓ Used between BMS, inverter, charger, VCU and dashboard.

CAN Message Example

Signal	Meaning	Used by
Battery SOC	Remaining battery percentage	Dashboard, EMS
Pack current	Charge/discharge current	BMS, inverter
Motor speed	RPM feedback	VCU, traction control
Fault status	Error or protection warning	All controllers

Energy Management Strategies

Rule-Based Control

Fixed rules such as engine ON below SOC limit, regenerative braking when brake is applied, motor assist during acceleration. Simple and reliable.

Optimization-Based Control

Minimizes fuel or energy using mathematical optimization. Better performance but needs computation and model accuracy.

Predictive Control

Uses route, traffic, gradient and driver demand prediction. Good for smart vehicles but requires data.

Fuzzy/AI Control

Handles nonlinear vehicle behaviour and uncertain driving patterns. Implementation must be validated carefully.

Implementation Issues

- ✓ Battery ageing and temperature variation.
- ✓ Sensor errors and communication delay.
- ✓ Driver behaviour is unpredictable.
- ✓ Regeneration limited by SOC and road condition.
- ✓ Safety limits override energy-saving commands.

Charging Techniques

Technique	Explanation	Important point
Slow AC charging	On-board charger converts AC to DC.	Suitable for home/overnight charging.
Fast DC charging	External charger gives controlled DC through BMS limits.	High power; needs thermal and grid management.
Smart charging	Charging power/time controlled based on grid load, tariff, SOC and user need.	Reduces grid stress and charging cost.
Regenerative charging	Motor works as generator during braking.	Limited by battery SOC and traction.

Formula Bank and Numerical Revision

Most calculation questions come from vehicle dynamics, energy, efficiency, range, power and torque.

Vehicle Dynamics

$$F_{rr} = C_{rr} mg\cos\theta$$

$$F_{ad} = 0.5\rho C_d A v^2$$

$$F_g = mg\sin\theta$$

$$F_a = ma$$

$P = Fv$

Energy, Battery and Range

Energy (Wh) = V × Ah

Usable energy = Nominal energy × depth of discharge

Range = usable Wh / Wh per km

Current = Power / Voltage

Charging time ≈ Battery energy / Charger power

Practice Question Bank

Short-answer, explanation, comparison, diagram and numerical styles.

Module 1

1 State the need for EVs.

2 Compare BEV, HEV, PHEV and FCEV.

3 Derive the total tractive force equation.

4 Explain rolling resistance, aerodynamic drag and gradient resistance.

5 Explain social and environmental importance of EVs.

Module 2

6 Draw and explain a basic electric drive train.

7 Compare major motor drives used in EV.

8 Explain regenerative braking power flow.

9 Compare series, parallel and series-parallel hybrid drive trains.

10 Explain fuel efficiency analysis of electric drive train.

Module 3

11 List battery parameters used in EV design.

12 Compare battery types.

13 Explain fuel-cell based energy storage.

14 Explain hybridization of battery and supercapacitor.

15 Describe motor and battery sizing.

Module 4

16 Explain the role of CAN in EV.

17 Classify energy management strategies.

18 Discuss EMS implementation issues.

19 Draw and explain a charging station schematic.

20 Explain smart charging.

Model Question Paper — 6042C

Original practice material based on the syllabus structure.

PART A — Short Answer Questions
Answer any ten. Each question carries 2 marks.
1. Define Battery Electric Vehicle.

2. What is regenerative braking?
3. List any four forces acting on a moving vehicle.
4. State the meaning of SOC and SOH.
5. What is a fuel cell electric vehicle?
6. List two advantages of permanent magnet motor in EV.
7. What is a series hybrid vehicle?
8. Define C-rate of a battery.
9. Mention two uses of CAN in electric vehicle.
10. What is smart charging?
11. List two charging station safety blocks.
12. State two limitations of solar powered vehicles.

PART B — Answer any five. Each carries 6 marks.
13. Explain EV operation with block diagram.
14. Derive total tractive force equation.
15. Compare electric drive train topologies.
16. Compare DC, induction, PM and SR motors.
17. Explain hybrid drive-train topologies.
18. Compare battery, fuel cell, supercapacitor and flywheel storage.
19. Explain CAN communication in EV.
20. Draw and explain charging station schematic.

PART C — Answer any two. Each carries 10 marks.
21. Explain types of EVs and their social importance.
22. Explain electric propulsion unit and drive system efficiency.
23. Explain drive system sizing procedure.
24. Classify energy management strategies and discuss implementation issues.

Numerical Practice
25. An EV consumes 140 Wh/km. Find battery energy needed for 100 km range if usable depth is 80%.
26. A vehicle requires tractive force of 500 N at 18 m/s. Find wheel power.

Answer Key and Revision Notes

Part A Short Answers

1. BEV uses battery and electric motor for propulsion without ICE.
2. Regenerative braking converts kinetic energy into electrical energy during braking.
3. Rolling resistance, aerodynamic drag, gradient force, acceleration force.
4. SOC is remaining charge percentage; SOH is battery health compared with new condition.
5. FCEV uses fuel cell, normally hydrogen based, to produce electricity for motor.
6. High efficiency and high power density.
7. In series hybrid, ICE drives generator and wheels are driven by electric motor.
8. C-rate is charge/discharge current relative to battery Ah capacity.
9. CAN carries BMS, motor controller, charger and dashboard messages.
10. Smart charging controls charging power/time based on grid, SOC and user requirement.
11. Protection, metering, isolation, charger controller, communication.
12. Low solar power density and dependence on sunlight/area.

Numerical Answers

25. Usable energy = 140 × 100 = 14000 Wh. Nominal energy = 14000 / 0.8 = 17500 Wh = 17.5 kWh.

26. Power = F × v = 500 × 18 = 9000 W = 9 kW.

High-Value Exam Points

✓ Always draw EV/HEV block diagrams with arrows.

✓ For motor comparison, write torque, efficiency, control complexity and maintenance.

✓ For batteries, mention energy density, power density, safety and BMS.

✓ For charging station, include grid, protection, metering, converter, communication and BMS.

✓ For CAN, write differential, priority-based, message-oriented communication.

References

Books

- James Larminie and John Lowry, Electric Vehicle Technology Explained, John Wiley & Sons.
- Mehrdad Ehsani, Y. Gao, S. E. Gay and A. Emadi, Modern Electric, Hybrid Electric, and Fuel Cell Vehicles, CRC Press.
- S. Onori, L. Serrao and G. Rizzoni, Hybrid Electric Vehicles: Energy Management Strategies, Springer.
- Iqbal Husain, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press.

- K. T. Chau, Electric Vehicle Machines and Drives, Wiley.

Study Method

Use online resources for deeper theory, but for exam preparation keep the syllabus sequence fixed: EV basics → drive train → storage/sizing → CAN/EMS/charging.

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